

Silent Decisions Are Type Errors: Enforcing AI Accountability via Linear Resource Types

Daniel Lau Pereira Soares Independent Researcher
São Paulo, Brazil
danniellau@gmail.com



Abstract—As autonomous AI agents make consequential decisions—credit, employment, healthcare—regulatory frameworks such as Brazil's LGPD (Art. 20) and the EU AI Act (Arts. 14, 86) demand auditable accountability. Policy-as-Code tools treat accountability as an asynchronous runtime property, leaving decisions able to exist without evidence: *silent decisions*. We introduce *Bound-To-Verdict* (BTV), shifting AI accountability from runtime audit obligation to compile-time type guarantee. BTV encodes regulatory obligations as linear resource types: a *Verdict* (V) cannot be constructed within Safe Rust without atomically consuming an *EvidenceToken* (E) and an authority-signed *ComplianceToken* (C), formalised as $(E \otimes C_{\text{signed}}) \multimap V$. We prove the *Constitutional Enclosure Theorem*: under the assumptions of Section 4.3, the public Safe Rust API cannot construct a *Verdict* without both tokens. A reference implementation shows the enforcement adds no runtime branch; the measured cost is that of the cryptographic primitives and durable persistence.

Index Terms—linear types, AI accountability, type systems, silent decisions, Policy-as-Code, Rust, LGPD, EU AI Act, fail-secure, formal verification

1 INTRODUCTION

Deployed AI systems now issue consequential decisions in milliseconds: a credit application is rejected; a social-benefit disbursement is blocked. Regulatory frameworks acknowledge this reality. Brazil's Lei Geral de Proteção de Dados (LGPD), Article 20, grants data subjects the right to request review of decisions based solely on automated processing that affect their interests; on request, controllers must provide clear and adequate information about the criteria and procedures, subject to protection of trade and industrial secrets [1]. The European Union AI Act requires human-oversight capabilities for high-risk systems (Article 14) and, in defined cases, a right to explanation (Article 86) [2]. Neither instrument, by itself, prevents software from emitting a consequential decision before the required evidence or review support exists: a system may emit a denial, log it asynchronously, and silently drop the log; the violation surfaces only later, if at all. We call this phenomenon a *silent decision*: an AI-generated act with legal consequences for an individual that carries no integrity-authenticated evidence

record (Figure 1). Silent decisions are not a failure of intent—they are a failure of type.

The conventional response is instrumentation (Figure 2). Policy engines such as Open Policy Agent [3] and Kyverno govern cloud infrastructure—pods, secrets, network policies—issuing allow/deny verdicts over *resources*, not over *persons*. Evidence of a decision's basis is a separable artifact, appended asynchronously, omissible under load, and falsifiable after the fact. Runtime assertions offer a second line of defense: `assert(evidence != null)` at every callsite. In C, `free()` can be paired with discipline and checked by analyzers—yet dangling-pointer vulnerabilities accumulated for four decades because discipline fails at scale. Rust's ownership model solved it by making the *absence of a deallocation owner* a compile-time error; no existing framework makes the absence of evidence one.

Linear Logic [4], [5] treats propositions as *resources*: used exactly once, neither duplicated nor discarded. The connective $\multimap$ (linear implication) denotes a resource transformation that *consumes* its premise; $\otimes$ denotes the *joint* consumption of two resources. We use the type law $(E \otimes C) \multimap V$: an *EvidenceToken* (E) and a *ComplianceToken* (C) are jointly consumed to produce a *Verdict* (V). Rust's ownership model is affine rather than strictly linear; Section 3 states the resulting limits explicitly. In our implementation, *EvidenceToken* is a `#[must_use]` struct with no `Clone` or `Copy` derivation; `Verdict::new()` takes both tokens by value, consuming them atomically. Because `Blake3Hash`—the only evidence artifact—has no public constructor, it cannot be forged: the sole path to a `Blake3Hash` is through `EvidenceToken::consume()`. A silent decision is therefore not an illegal act that slips past a guard; it is a *type error*: the compiler rejects it before a single byte of evidence is ever dropped.

This paper makes three contributions.

- 1) **Linear Resource Types for AI Accountability (C1).** We instantiate the linear resource discipline established for protocols and resources (Vault, typestate, session types; Section 2.3) over AI accountability obligations—LGPD Art. 20 and EU AI Act Arts. 14

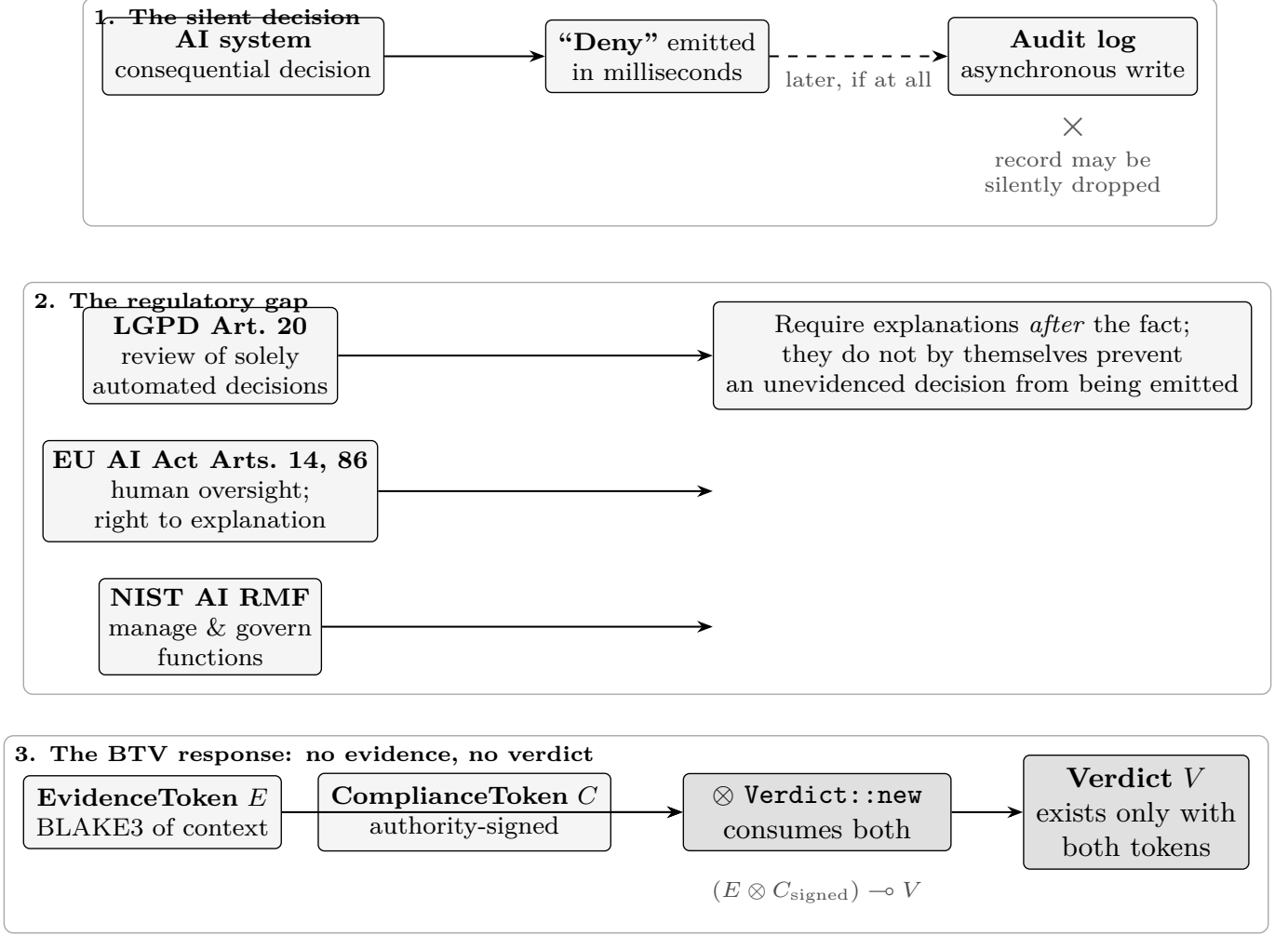


Fig. 1. The silent-decision problem and the BTV response. Top: a consequential decision is emitted and its audit record, written asynchronously, may be lost. Middle: LGPD Art. 20 and EU AI Act Arts. 14 and 86 require review and explanation but do not by themselves prevent unevidenced decisions. Bottom: BTV makes evidence and an authority-signed compliance token preconditions of verdict construction.

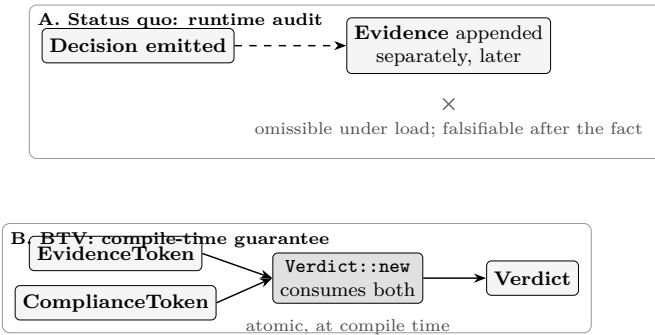


Fig. 2. The status quo (top) versus the BTV guarantee (bottom): evidence is either a separable, omissible artifact appended after the decision, or a precondition consumed atomically by `Verdict::new`.

and 86: the core invariant $(E \otimes C_{\text{signed}}) \multimap V$ is a compile-time type constraint in Rust, verified on the companion repositories [6], [7].

- 2) **Decision-Evidence Causal Binding (C2).** We prove

that, under the *BTV*¹ type invariant, every materialized `Verdict` consumed *exactly one* causally-prior `EvidenceToken`: a BLAKE3 digest of the decision context computed before the verdict existed. Linear ownership prevents duplication; the private constructor prevents forgery; both guarantees are static.

- 3) **The Constitutional Enclosure Theorem (C3).** We state and prove that external Safe Rust callers cannot construct a BTV `Verdict` through the public API without both required tokens. The proof analyzes the crate’s public construction paths and is machine-checkable: `cargo test` on the companion repository verifies all seventeen clauses in seconds.

2 RELATED WORK

Algorithmic accountability intersects infrastructure governance, runtime verification, and type theory. We situate our

1. BTV (*Bound-To-Verdict*): a reference kernel implementing atomic accountability via linear types; the name restates the type law $(E \otimes C) \multimap V$.

contribution against the prevailing methodology in each domain.

2.1 Policy-as-Code for Infrastructure Governance

The industry standard for continuous compliance relies on Policy-as-Code (PaC) frameworks such as Open Policy Agent [3], Kyverno, and Checkov. These tools excel at governing infrastructure artifacts—that a Kubernetes pod lacks privilege escalation, or that a Terraform plan avoids over-privileged IAM roles. However, they are designed to evaluate *declarative configurations*, not *dynamic runtime decisions* affecting individual rights.

Evidence of compliance manifests as asynchronous audit logs: records written after the decision logic executes, stored separately, and subject to omission or alteration if the logging pipeline fails. PaC frameworks govern configuration artifacts, not transient decision acts with legal consequences for a person; accountability remains a post-hoc state, structurally vulnerable to the silent-decision failure mode of Section 1.

2.2 Runtime Assertions and Audit Logging

To govern application-level decision logic, practitioners typically rely on a combination of runtime assertions (e.g., `assert(evidence != null)`) and structured audit logging. While functional under nominal conditions, these are procedural promises, not structural invariants. Assertions can be disabled in production builds (stripping `debug_assertions` in Rust; omitting `-ea` in the Java VM); audit logs can be truncated under load, dropped by a misconfigured sink, or bypassed by an untested code path.

Both mechanisms operate *after* the decision logic executes: they do not prevent a verdict that lacks integrity-authenticated evidence from materializing; they attempt to record its absence, if they run at all. This architectural decoupling separates the act of deciding from the act of preserving evidence needed to support review, explanation, and oversight under LGPD Art. 20 and EU AI Act Arts. 14 and 86.

A stronger form of runtime guarantee is offered by transactional databases: an ACID transaction wrapping both the decision record and the audit log ensures that either both are committed or neither is. However, transactional atomicity is a tool the developer must consciously choose to apply—wrapping decision and log within the same `BEGIN/COMMIT` boundary. Omitting that boundary is a logic error, not a type error: the compiler accepts the program, the database commits the decision without the log, and the silent-decision failure mode materialises with no static diagnostic. BTV closes exactly this gap: the compiler rejects any program that constructs a `Verdict` without a causally-prior `EvidenceToken`, regardless of whether the developer intended to include the evidence or not.

2.3 Linear Types and Resource Correctness

Linear Logic’s structural resource guarantees [4], [5] were adapted into programming-language design and underpin Rust’s ownership and borrowing model [8]. Prior applications of linear and substructural types, however, focus predominantly on *resource correctness*: ensuring that memory

is freed exactly once, that file handles are not leaked, or that a communication protocol is followed in order. Vault enforces high-level resource-management protocols in low-level code via linear typestates [9]; typestate systems make state-dependent protocol obligations explicit at the type level [10]; session types impose communication protocols on channels; and the substructural design space of affine versus linear versus relevant systems is surveyed by Walker [11], with state-dependent disciplines studied by Ahmed et al. [12]. Our claim C1 is deliberately narrower: we *instantiate* that established discipline over *regulatory-support* artifacts—a BTV verdict must consume causally bound evidence and an authority-issued compliance token—and relate the mechanism to LGPD Art. 20 and EU AI Act Arts. 14 and 86. We scope the resulting guarantee to the Safe Rust type perimeter (Section 3.3), with the escape hatches of Section 3 stated explicitly. Bound-To-Verdict extends the linear-type discipline from resource management to regulatory accountability, not from nothing to regulatory accountability.

3 A LINEAR TYPE SYSTEM FOR AI ACCOUNTABILITY

The accountability gap of Section 1 stems from the mismatch between *sequential* runtime assertions and *concurrent, high-throughput* AI decision pipelines. This section introduces the linear-type framework that closes the gap at the language level.

3.1 Regulatory Obligations as Linear Resources

Girard’s linear logic [4] treats propositions as *resources*: each assumption must be used exactly once—no duplication (*contraction*) and no silent discard (*weakening*). This discipline maps directly onto regulatory accountability obligations.

To support review and explanation under LGPD Art. 20 and EU AI Act Arts. 14 and 86, BTV models two accountability artifacts as resources. If an evidence resource could be duplicated, the same evidence could be recycled across multiple decisions, undermining causal uniqueness. If the obligation could be silently discarded, decisions could escape accountability entirely—the precise failure mode we call a *silent decision*. We model the two obligations as first-class linear types:

EvidenceToken A linear resource that encapsulates the BLAKE3 hash of the decision context (the raw input fed to the AI system at inference time). Construction via `EvidenceToken::new(&[u8])` binds the token irrevocably to a specific context. The type does not implement `Clone` or `Copy`, enforcing an “at most once” discipline at the type-system level.

ComplianceFramework A linear resource that encodes the applicable regulatory framework: jurisdiction identifier, policy version, and contestability deadline in hours. It is consumed by value, binding each regulatory obligation to exactly one verdict.

The structural guarantee is narrower than non-repudiation: because an `EvidenceToken` cannot be cloned through the public API, one token cannot be consumed

$$(E \otimes C) \multimap V$$

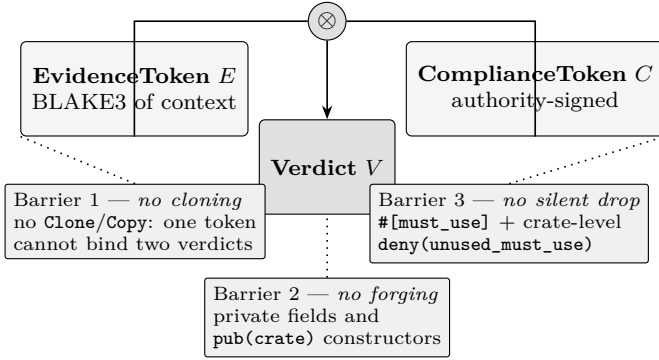


Fig. 3. The core type law $(E \otimes C) \multimap V$: an EvidenceToken and a ComplianceToken are jointly consumed ($\otimes$) to construct a Verdict; compiler-enforced barriers block cloning and unauthorized construction, and diagnose an ignored token-producing expression at the API boundary.

by two verdict constructions. Tokens can still be dropped because Rust ownership is affine; what remains mandatory is that every successful BTV Verdict construction receives both tokens. We return to the affine-versus-linear distinction below (Section 3.2.0.1).

3.2 The Atomic Binding Operator

The central type law of this paper is:

$$(E \otimes C) \multimap V$$

where V denotes Verdict, E denotes EvidenceToken, C denotes ComplianceToken, $\multimap$ is the linear function type (“linear implication”), and $\otimes$ is the multiplicative conjunction (“tensor product”) of linear logic.

Reading left-to-right: to produce a V , one must jointly consume exactly one E and exactly one C . There is no partial construction; either both resources are provided and a verdict is produced, or the program does not type-check (Figure 3).

In the implementation, `Verdict::new` is the concrete realisation of the $\otimes$ -introduction rule (Axiom 4.3):

```
pub fn new(
    evidence: EvidenceToken,
    compliance: ComplianceToken,
    decision: Decision,
    explanation: String,
) -> Verdict
```

Both are taken *by value*: after the call, neither token is accessible to the caller—they have been irreversibly consumed into Verdict. The HMAC integrity seal computed inside `new` binds `evidence_id`, `decision`, and `explanation` into a tamper-evident record, making post-hoc alteration detectable.

3.2.0.1 Affine, not strictly linear. Precision matters: in substructural-type-theory terms [11], [10], Rust ownership is *affine* (used zero or one times) rather than strictly *linear* (exactly once) — a value can always be silently dropped. `#[must_use]` on

TABLE 1

The three protections of the Constitutional Enclosure. Attack vectors, left to right: constructing a Verdict by struct literal; calling `EvidenceToken::consume()` from outside the crate; and dropping an EvidenceToken without consuming it.

ID	Attack vector	Mechanism	Error
A	Struct literal	Private fields	E0451
B	External <code>consume()</code>	<code>pub(crate)</code>	E0624
C	Ignored return value	<code>#[must_use]</code>	<code>unused_must_use</code>

EvidenceToken and ComplianceToken, plus a crate-level `#![deny(unused_must_use)]` (Protection C, Table 1), escalates an ignored-return warning to a hard error, but does not turn Rust into a strictly linear language. Three escape hatches remain, documented rather than hidden: `std::mem::forget(token)` still compiles and leaks the token; a `panic!` after construction but before consumption drops it during unwinding; a process abort loses it with the process. None lets a Verdict materialize without evidence — each only discards evidence, a fail-secure outcome, not a silent decision (Definition 4.2) — but they mark the exact boundary of the “exactly once” guarantee.

This property is stronger than a call-site runtime assertion, which executes only if the call site is reached and can be omitted from an untested path; the ownership constraint is enforced by the type checker and applies to every compilation and every code path.

3.3 Encapsulation as Compile-Time Enforcement Boundary

The type law alone is insufficient if an adversary (or an inattentive developer) can construct a Verdict by other means—for instance, by writing a struct literal or by directly assembling the hash value. Three complementary protections close the encapsulation perimeter:

Protection A (Table 1) prevents assembling a Verdict without the constructor, regardless of the caller’s knowledge of the field types. Protection B prevents extracting the inner Blake3Hash for reuse: the `pub(crate)` qualifier restricts `consume()` to the defining crate, and external calls are rejected at compile time. Protection C turns token discard into a hard error: `#[must_use]` combined with a crate-level `#![deny(unused_must_use)]`.

Together, these three protections form a *closed perimeter*: the only well-typed path to a Verdict in any external crate is through `Verdict::new`, which enforces the $(E \otimes C) \multimap V$ law.

3.3.0.1 Scope of the Theorem. The Constitutional Enclosure Theorem (Section 4) holds within the following boundaries:

- **In scope:** All Safe Rust code. The type checker enforces the visibility and ownership rules without exception.
- **Out of scope:** Code using the `unsafe` keyword. `unsafe` permits raw pointer manipulation and can bypass visibility rules. Deployments should enforce a no-unsafe policy in first-party crates and audit dependencies; cargo geiger inventories unsafe usage but does not prove soundness.

- **Out of scope:** Semantic truthfulness of the evidence. The theorem guarantees that *some* context was hashed, not that the context is the correct or complete input. Policy-level correctness is a separate concern addressed at the application layer.
- **Out of scope:** Policy accuracy. Whether the `ComplianceToken` accurately reflects the applicable law is a legal question, not a type question.
- **Out of scope:** Decisions that bypass the `Verdict` type entirely. The theorem guarantees that no `Verdict` exists without evidence—it does not guarantee that all consequential decisions are routed through `Verdict`. The pattern `let _ = EvidenceToken::new(...)` silences the `#[must_use]` lint but produces no `Verdict`, so it is not a silent decision under Definition 4.2; routing all decision points through BTV is an architectural, not type-level, obligation.

These boundaries reflect a deliberate separation of concerns: the public API enforces the *structural* property that a `Verdict` cannot be constructed without the required resources; *substantive* accountability—accurate evidence, correct policy—belongs to audit, certification, and human review.

4 THE CONSTITUTIONAL ENCLOSURE THEOREM

This section formalises the paper’s bounded claim: an external Safe Rust caller cannot obtain a BTV `Verdict` through the crate’s public API without supplying the required evidence and compliance tokens. This is an API-enclosure result, not a theorem that every consequential action in a surrounding system is routed through BTV.

4.1 Definitions

Definition 4.1 (Materialized Verdict). *A verdict V is materialized if and only if there exist an evidence token E , a compliance token C , a decision outcome $d \in \{\text{Allow}, \text{Deny}\}$, and an explanation string x such that*

$$V = \text{Verdict}::\text{new}(E, C, d, x).$$

The function `Verdict::new` is the sole constructor of the type `Verdict`; no other term of that type is well-typed in Safe Rust.

Definition 4.2 (Silent Decision). *For a program P , let S_P denote the set of unevidenced BTV verdicts: terms of type `Verdict` exposed through the crate’s public API without construction from one `EvidenceToken` and one authority-issued `ComplianceToken`. This definition is deliberately confined to the BTV type perimeter; an external action that bypasses `Verdict` is outside the set.*

4.2 Axioms

Axiom 4.3 (Resource Introduction, $\otimes$ -I). *If $\Gamma \vdash E : \text{EvidenceToken}$ and $\Delta \vdash C : \text{ComplianceToken}$, then*

$$\Gamma, \Delta \vdash \text{Verdict}::\text{new}(E, C, d, x) : \text{Verdict}.$$

The contexts Γ and Δ are disjoint: linear logic forbids sharing a resource between both premises. In Rust this is enforced by the move semantics of `Verdict::new`, which takes E and C by

value. Once passed to the constructor, neither token is accessible to the caller.

Axiom 4.4 (Resource Uniqueness and Drop Diagnostics). *`EvidenceToken` is annotated `#[must_use]` and does not implement `Clone` or `Copy`. Consequently:*

- 1) *Contraction is prohibited: a token cannot be duplicated, preventing the same evidence from being associated with more than one verdict.*
- 2) *Ignoring a token-producing expression emits an `unused_must_use` diagnostic, treated as an error in this crate. Explicit discard remains possible because Rust is affine; this lint is a guardrail, not an anti-weakening proof.*

The method `consume()` is the unique destructor; it transfers the inner `Blake3Hash` to the `Verdict` constructor and is declared `pub(crate)`, making it inaccessible outside the defining crate.

Axiom 4.5 (Encapsulation Boundary). *Let Γ_{ext} denote any typing context external to the defining crate (i.e., any context in a crate that depends on `silent-decisions-proof` but is not the crate itself). The following are ill-typed in Γ_{ext} :*

- 1) *Struct-literal construction of `Verdict`—all fields are private (compiler error E0451).*
- 2) *Direct invocation of `EvidenceToken::consume()`: the method is `pub(crate)` (compiler error E0624).*
- 3) *Construction of `Blake3Hash` via its tuple-struct field—the field is private (compiler error E0423).*

Together, these restrictions close the encapsulation perimeter.

4.3 Threat Model

The theorem holds under the *Safe Rust* memory model. The `unsafe` keyword grants the programmer the ability to bypass borrow-checker and visibility rules; consequently, code that uses `unsafe` is explicitly out of scope.

First-party crates forbid `unsafe`; production deployments should also inventory dependency usage with `cargo geiger` [13]. That tool reports `unsafe` constructs but does not prove their soundness or guarantee that the theorem composes across the entire dependency graph.

Macro-generated code is in scope: procedural macros run before the type checker and cannot produce terms that circumvent visibility rules without themselves using `unsafe`.

4.4 Proof of the Constitutional Enclosure Theorem

Theorem 4.6 (Constitutional Enclosure). *Assume the BTV crate implementation and its public encapsulation are preserved. No external Safe Rust caller can obtain a term of type `Verdict` through that public API without the constructor consuming one `EvidenceToken` and one authority-issued `ComplianceToken`. Equivalently, $S_P = \emptyset$ within this API perimeter.*

Proof. We analyze the public construction and extraction paths closed by the crate’s encapsulation assumptions (Figure 4).

Case A: Struct-literal construction. An adversary attempts to write

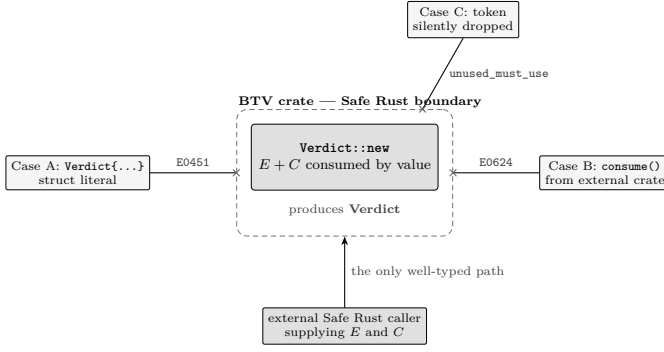


Fig. 4. The Constitutional Enclosure perimeter: attack vectors A–C are rejected by the compiler (E0451, E0624, unused_must_use); the only well-typed path to a Verdict is Verdict::new.

```
Verdict { decision: Decision::Allow, .. }
```

from a crate external to silent-decisions-proof. By Axiom 4.5(1), all fields of Verdict are private. The compiler rejects this expression with error[E0451]: field ‘evidence_id’ of struct ‘Verdict’ is private. This is verified by the compile-fail test tests/ui/verdict_struct_literal.rs.

Case B: External invocation of consume(). An adversary attempts to call EvidenceToken::consume() directly from an external crate, bypassing Verdict::new to extract a Blake3Hash and assemble a Verdict fragment. By Axiom 4.5(2), consume() is declared pub(crate). The compiler rejects the call with error[E0624]: method ‘consume’ is private. This is verified by the compile-fail test tests/ui/external_consume_call.rs.

Guardrail C: Discard of EvidenceToken. A developer issues a decision without providing evidence by constructing an EvidenceToken and immediately dropping it (or ignoring the return value of a function that returns one). By Axiom 4.4, ignoring a token-producing expression triggers a diagnostic promoted to an error in this crate. Explicit let _ = expr, drop, or mem::forget remains possible. None produces a Verdict; the formal closure instead comes from Axiom 4.5 (private fields and restricted visibility). This is verified by the compile-fail test tests/ui/dropped_evidence_token.rs.

By the Encapsulation Boundary axiom, no term of type Verdict can be obtained in Γ_{ext} without passing through Verdict::new. By Definition 4.1, every such term is a Materialized Verdict; therefore $\mathcal{S}_P = \emptyset$ within the stated perimeter. $\square$

4.5 Corollary: Fail-Secure by Construction

Corollary 4.7 (Fail-Secure). *In any Safe Rust program satisfying the conditions of Theorem 4.6, the code path that handles a runtime failure to produce evidence (e.g., a database timeout, a missing context field) cannot silently issue a Verdict—the type system ensures at compile time that every branch leading to a Verdict must supply a valid EvidenceToken. The protected path fails securely: it cannot return a BTV verdict that lacks an evidence record.*

Proof. The only well-typed continuation that produces a Verdict requires unwrapping the result and handling the error branch explicitly; that branch must either propagate the error or halt. By Theorem 4.6, any attempt to construct a Verdict in the error branch without a valid EvidenceToken is ill-typed. $\square$

The Fail-Secure Corollary supports review and explanation under LGPD Art. 20 and EU AI Act Arts. 14 and 86 by making the required artifacts structural on the protected BTV path rather than an optional post-hoc assertion.

4.6 Corollary 4.8: Human Escalation as the Only Well-Typed Alternative

Corollary 4.8 (Human Escalation). *Under the conditions of Theorem 4.6, when an EvidenceToken cannot be produced (e.g., model timeout, unavailable context), the API provides a well-typed alternative to error propagation or halt: constructing an EscalatedVerdict via a linearly-consumed OperatorToken. Silent fallback to an unevidenced automated verdict is ill-typed. Formally, $O \multimap V_{esc}$.*

This structure aligns with the human oversight requirement of EU AI Act Art. 14: the system cannot produce a consequential decision-equivalent record without binding exactly one named human operator, whose identity and authorisation are protected by an HMAC integrity seal under the configured key-management model.

Proof. EscalatedVerdict is a sibling type to Verdict and does not affect the proof of Theorem 4.6. std::mem::size_of::<EscalatedVerdict>() measures 160 bytes versus 184 for Verdict (x86-64, rustc 1.98.1; Section 5.2) — the automated verdict is the larger type because it carries the signed compliance token’s 32-byte signature. Its sole constructor EscalatedVerdict::new requires an OperatorToken by value, following the same $\otimes$ -I pattern as Axiom 4.3. OperatorToken carries the same linear-resource invariants as EvidenceToken: it is #[must_use], not Clone/Copy, and its consume() destructor is pub(crate). The four compile-fail tests escalated_struct_literal.rs, escalated_token_reuse.rs, escalated_operator_token_drop.rs, and escalated_consume_external.rs exercise the corresponding encapsulation and ownership cases of Theorem 4.6; no EscalatedVerdict can be forged, duplicated, or silently dropped. $\square$

4.7 Polyglot Deployment: The FFI Trust Boundary

Most production ML pipelines are polyglot: orchestration logic (agent frameworks, model servers, API gateways) runs in Python and other languages, while BTV’s linear-type discipline is enforced by the Rust compiler. The PyO3 gateway (btv-python) is the sole authorized boundary between these worlds: a Python caller can invoke issue_verdict, but cannot inspect, clone, or reconstruct a Verdict—it receives only an opaque, integrity-sealed SealedVerdict handle exposing read-only accessors and a verify_integrity check. Theorem 4.6 is therefore enforced strictly within the btv-core and btv-python crates; it does not, and cannot, extend to the untrusted

orchestrator itself. Consequently, the enforcement burden shifts downstream: the effector that materialises a decision’s real-world consequence (the credit-denial API, the database write that blocks a transaction) must independently require and verify the sealed verdict handle before acting, and must reject any request that bypasses the gateway. The Rust enclave protects itself; it does not, by itself, protect the system that surrounds it.

5 EMPIRICAL VALIDATION

The Constitutional Enclosure Theorem guarantees that an external Safe Rust caller cannot construct a BTV `Verdict` through the public API without the required tokens. Does that guarantee impose a runtime cost? We answer across payload sizes, thread counts, and two persistence postures, in the direction the measurements point—which is not the flattering one.

5.1 Experimental Setup

5.1.0.1 Methodology.: Micro-benchmarks use Criterion’s `iter_batched` discipline with `black_box` on every input crossing an optimization boundary and a per-iteration seed making each hashed context distinct. The sweep harness’s hot path carries no injected synchronization; percentiles are estimated online with the P^2 algorithm [14] and are therefore *estimates*, not order statistics. Each configuration runs five trials; we report medians and the trial-to-trial coefficient of variation (CV). Iteration counts are calibrated per (payload, mode) using that mode’s own operation, so they are not comparable across modes.

5.1.0.2 Hardware, software, and what this collection is not.: All three tables come from one collection on one host: an Intel Xeon virtual machine, 4 vCPUs, Linux x86_64, `rustc 1.98.1` pinned via `rust-toolchain.toml`, 5 s wall target per configuration. Two constraints bound it. The durable arm writes to a containerized overlay filesystem, so its `fsync` cost is a *lower* bound on bare-metal storage; and this is a reduced-footprint snapshot—the 90 s-per-configuration run on dedicated hardware, and a second-platform collection with the current code, remain open obligations in the artifact’s evidence manifest. We publish it with its environment fingerprint because it is reproducible today and internally consistent, not as a headline result. `aarch64` coverage is compile-and-test only and carries no performance claim.

5.2 Results

5.2.0.1 Construction latency.: Table 2 reports the end-to-end cost of `Verdict::new`—BLAKE3 context hashing, authority-signature verification, HMAC-SHA256 sealing, and allocation. The payload-scaling portion of the curve is BLAKE3; the $\approx 5.36 \mu\text{s}$ floor is dominated by two HMAC-SHA256 computations—verifying the compliance token’s authority signature, then sealing the verdict—plus hex encoding of the evidence identifier. Each is work an accountable system must perform whatever its governance architecture; none is attributable to the type discipline, which emits no instruction at all.

TABLE 2

Latency of `Verdict::new` by context size (Criterion 95% CI; `rustc 1.98.1`). Includes authority-signed token issuance and verification. Provenance: `data/construction_latency_criterion.csv`, generated by `scripts/gen_construction_table.py` from `btv-core/target/criterion` (regenerate with `cargo bench --features test-support --bench verdict_construction -- verdict_construction --save-baseline construction_provenance`).

Context size	Lower	Mean	Upper
64B	5.331 μs	5.359 μs	5.405 μs
512B	5.808 μs	5.816 μs	5.825 μs
4KiB	7.621 μs	7.650 μs	7.682 μs

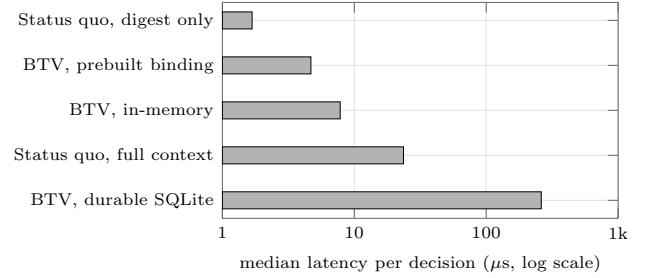


Fig. 5. Measured median latency of the BTV and status-quo logging paths at 4 KiB, one thread (logarithmic scale). The rows compare distinct implementation and persistence postures, not a single end-to-end number; type-level enforcement adds no runtime branch, and the measured costs are those of hashing, token verification, sealing, encoding, and—where enabled—durable persistence.

5.2.0.2 Thread scaling, and where contention actually lives.: Table 3 separates the two persistence postures under concurrency. The in-memory path is flat—7.82 μs at one thread and at four, CV below 0.3%—the empirical signature of the absent lock contention claimed in Section 3: the linear discipline introduces no shared state to contend for. The durable path inverts that. Its median degrades mildly (261 to 292 μs) while its p99 rises from 512 μs to 28 ms as four threads serialize on one log connection. That tail belongs to the chosen sink, not the type system, and it is the honest answer to whether this design ships production-ready: the reference `LogSink` is a single-writer SQLite file, and scale needs a sink built for concurrent append. The type-level guarantee is indifferent to that choice; the latency budget is not.

5.2.0.3 Why the construction rows differ.: Table 2 times the full public path, including token issuance and verification. Table 4’s smaller “prebuilt-token binding” row creates tokens outside the timed region and measures only their binding. It is not a second end-to-end estimate.

5.2.0.4 What accountability costs against the status quo.: Table 4 and Figure 5 show the comparison an architect actually faces. Against a status quo that logs the complete decision context, the durable BTV pipeline is $11.1\times$ slower at 4 KiB; against one that logs only a digest and metadata, $155.5\times$ slower. We report both because which one counts as “the status quo” is a modelling choice rather than a fact about BTV, and we state the direction plainly: an earlier revision of this section claimed $2.6\times$ faster against the full-context baseline, a figure produced by a benchmark defect described below.

TABLE 3

Thread scaling at 4 KiB payloads, in-memory versus durable persistence. Each cell is the MEDIAN across five trials of the per-trial aggregate; CV% is the trial-to-trial coefficient of variation. Percentiles are P^2 estimates (Jain & Chlamtac 1985), not order statistics. Provenance: data/sweep_raw_20260915T024430Z_g5fix.csv (Intel Xeon, 4 vCPU, 5 s target).

Threads	In-memory sink			Durable SQLite (WAL, FULL)		
	p50 (μs)	p99 (μs)	CV%	p50 (μs)	p99 (μs)	CV%
1	7.82	25.87	0.1	261.44	511.55	7.8
2	7.83	24.14	0.3	301.15	10929.66	2.0
4	7.82	24.50	0.2	291.85	28000.83	8.8

TABLE 4

Five-mode accountability contrast at 4 KiB payloads, one thread (Xeon 4 vCPU, 5 s target; reduced-footprint snapshot). Each cell is the median across five trials; percentiles are P^2 estimates. The durable arm uses a real on-disk SQLite log (WAL, `synchronous=FULL`); its container-storage numbers are a LOWER bound on bare-metal fsync cost. Every durable-mode operation issued a real row (end-of-run sanity gate: rows persisted == operations issued), closing a prior round's silent idempotent-replay defect (COMSI-2026-04-0112 Round 2, G5). Provenance: data/sweep_raw_20260915T024430Z_g5fix.csv. Rows: BTV in-memory sink; BTV with durable on-disk SQLite; BTV binding with tokens built outside the timed region; status quo digest plus metadata; status quo logging the full context as JSON, fire-and-forget.

Mode	p50 (μs)	p99 (μs)	throughput (k ops/s)
BTV, in-memory sink	7.82	25.87	113.9
BTV, durable SQLite	261.44	511.55	3.1
BTV, prebuilt binding	4.69	8.56	114.7
Status quo, digest only	1.68	2.28	507.6
Status quo, full context	23.61	46.04	38.4

The multiplier is real, and so is what the baseline buys with it. The fire-and-forget arm is fast partly because it is permitted to lose the record: the artifact's contrast experiment (`tests/test_status_quo_contrast.rs`) shows dead and backlogged loggers each returning success for a thousand decisions while every record is discarded—the silent decision of Section 1 reproduced under laboratory conditions. BTV's durable arm returns a `Verdict` only after the row exists. The comparison is therefore not like-for-like, and we decline to present it as though it were: what the numbers price is the difference between a record that exists and one that might.

5.2.0.5 Correction of record.: These figures replace earlier ones that were wrong: the harness seeded each payload from a trial-local counter, so later trials replayed evidence the log already held, and the append-only sink correctly returned success without a second write. Four of every five trials measured that no-op. Every durable payload now draws from one process-wide atomic counter, and each run asserts that rows persisted equals operations issued—this collection passed exactly, 1,803,460 of 1,803,460. We record the correction in the manuscript because a defect that makes the accountable path look cheaper than it is defeats this work's purpose.

5.3 Worked Examples

Three worked examples demonstrate BTV in semi-realistic scenarios: credit scoring under LGPD (30-day appeal window), hiring screening under EU AI Act Art. 86, and emergency triage (where Allow decisions also require evidence); source code is in the companion repository (`examples/`).

6 DISCUSSION

The Constitutional Enclosure Theorem establishes a strong structural guarantee within a precisely bounded scope. This section examines the boundaries of Theorem 4.6, identifies

limitations of the reference implementation, and proposes extensions that close each gap without weakening the core invariant.

6.1 Scope and Boundaries of the Guarantee

6.1.0.1 Unsafe Rust.: The scope limits of §3.3 and §4.3 apply here too: enforcing a zero-unsafe policy across the dependency graph (auditable via `cargo geiger` [13]) is an operational decision, not a theoretical barrier, and requires no change to the BTV crate itself.

6.1.0.2 Semantic truthfulness and the AttestableContext extension.: The theorem guarantees that *some* byte sequence was hashed and bound to a verdict, not that it faithfully represents the context presented at inference time: a caller could pass `b"placeholder"` to `EvidenceToken::new` and obtain a structurally valid token. A sealed `AttestableContext` trait would narrow this gap without weakening the core invariant:

Listing 1. AttestableContext trait (proposed extension)

```
1 pub trait AttestableContext: Sealed {
2     fn canonical_bytes(&self) -> &[u8];
3     fn schema_version(&self) -> u16;
4     fn source_attestation(&self) -> SourceAttestation;
5 }
```

where `SourceAttestation` carries a signature over the payload from an HSM/TPM-backed key; `EvidenceToken::new` would accept `impl AttestableContext` instead of `&[u8]`, reducing forgery to compromising that key rather than passing an arbitrary byte string. This tightens the type law to $(E_{\text{attested}} \otimes C) \multimap V$ —a stronger premise than $(E \otimes C) \multimap V$ —while the `Sealed` supertrait keeps external crates from implementing `AttestableContext`, preserving Axiom 3's closed perimeter.

6.2 Limitations of the Reference Implementation

The `silent-decisions-proof` kernel is a proof-of-concept designed to validate the theoretical claims of this paper. Four implementation choices trade production robustness for clarity of exposition; each has a well-defined remediation path.

6.2.0.1 L1 — Static HMAC key (mitigated): The reference implementation reads the HMAC key from `BTV_HMAC_KEY` at runtime, falling back to a compile-time constant only when unset, closing L1 for deployments that inject a key from an HSM/KMS. This surfaces a recurring asymmetry: the verdict seal’s integrity and authenticity claim holds only when `BTV_HMAC_KEY` shares HSM/KMS infrastructure with the audit log’s own authentication key. HMAC alone does not provide third-party non-repudiation. The type-level guarantee (Theorem 4.6) is independent of key management and holds regardless of the key source.

6.2.0.2 L2 — ComplianceAuthority factory, HMAC-signed (implemented): The `ComplianceToken::new` constructor was originally `pub`, letting any external crate self-declare a jurisdiction without validation or authority. The reference implementation restricts it to `pub(crate)` behind a `ComplianceAuthority` factory that validates the jurisdiction against an allowlist of recognised regimes (BR-LGPD, EU-GDPR, EU-AI-ACT) and signs (`jurisdiction`, `policy_version`, `deadline_hours`) with an HMAC key read from `BTV_AUTHORITY_KEY`. `Verdict::new` verifies that signature before constructing a `Verdict`, rejecting forged or differently-keyed tokens (`BtvError::InvalidTokenSignature`)—allowlist membership alone is no longer sufficient, closing the gap between “validated” and “authorized.” The type law tightens to $(E \otimes C_{\text{signed}}) \multimap V$; the encapsulation boundary (Axioms 1–3) is preserved.

6.2.0.3 L3 — Single-node evidence chain: `EvidenceToken` captures a single BLAKE3 hash; multi-step pipelines (retrieval, re-ranking, generation, scoring) each carry accountability implications. A *chain* of evidence tokens, each stage consuming the prior and producing a new one, generalises naturally while preserving $(E \otimes C) \multimap V$ at each step.

6.2.0.4 L4 — Polyglot pipelines: Since the theorem holds within Safe Rust (Axiom 4.5), external code is Γ_{ext} by definition and cannot construct a `Verdict` directly; §4.7 analyses this as a threat model, where the PyO3 gateway keeps external callers inside Γ_{ext} and a *BTV sidecar* per service could chain a fresh `EvidenceToken` at each stage, addressing L3 and L4 jointly.

6.3 CAL Design Trade-Offs

A production governance kernel faces a three-way tension among Consistency (every verdict carries a causally-bound evidence chain), Availability (the system always issues a verdict in finite time), and Legality (tokens come only from a verified authority). Existing audit-logging designs choose consistency-with-availability over legality: decisions keep issuing even when the logging pipeline fails, so the log carries no enforceable evidentiary guarantee. BTV chooses the opposite: it fails securely rather than issuing unevidenced

verdicts, accepting reduced availability, partially recovered by `EscalatedVerdict` (§4.6). This is a practical design observation, not a formal impossibility result; whether an FLP-style theorem [15] holds for accountability systems is open.

6.4 Future Work

Six directions are most immediate: (1) implementing `AttestableContext` (§6.1) to close the semantic-truthfulness gap; (2) extending the evidence chain to multi-step pipelines via the sidecar pattern (L3, L4) for retrieval-augmented and agentic architectures; (3) mechanising the Constitutional Enclosure Theorem in a proof assistant (Lean 4 or Coq), elevating it from a machine-checked Rust proof to a fully mechanised one; (4) testing whether a formal CAL impossibility result holds under asynchrony (§6.3); (5) unifying `Verdict` and `EscalatedVerdict` under `AccountableDecision` into the composed invariant $(V_{\text{auto}} \oplus V_{\text{esc}}) \multimap R$; and (6) composing BTV’s construction-time guarantee with tamper-evident persistence and privacy-preserving redaction, which we conjecture is tractable because the three operate at distinct layers with non-overlapping trust assumptions.

7 CONCLUSION

Regulatory frameworks such as LGPD Art. 20 and EU AI Act Arts. 14 and 86 establish review, information, explanation, and oversight obligations, but do not by themselves prevent software from emitting a decision without supporting evidence. We called this failure mode a *silent decision* and argued that its root cause is a failure of type: architectures treat accountability as a runtime audit obligation rather than a compile-time invariant.

Drawing on Girard’s Linear Logic [4] and Rust’s ownership model [8], we introduced the Bound-To-Verdict (BTV) framework and its core type law: $(E \otimes C) \multimap V$. Under this invariant, a `Verdict` cannot be constructed without atomically consuming an `EvidenceToken` and a `ComplianceToken`. The structural protections—private fields, `pub(crate)` consumption, and an affine ownership API reinforced by `#[must_use]` lint policy—enclose the authorized constructor. Within that perimeter, an unevidenced BTV verdict construction is rejected at compile time.

We proved the *Constitutional Enclosure Theorem*: under the assumptions of Section 4.3, an external Safe Rust caller cannot obtain a `Verdict` through the public API without the required tokens; the construction paths are machine-checked (cargo test: seventeen clauses), with a scoped Fail-Secure Corollary.

Empirical benchmarks on the reference implementation bound what the type-level enforcement itself costs: it adds no runtime branch. The measured cost is that of the cryptographic primitives and encoding (BLAKE3 + two HMAC-SHA256 computations, 5.36–7.65 μs for the full public construction path across 64 B–4 KiB payloads) and of durable persistence. On the reduced-footprint reference collection, durable SQLite reached a 261 μs median at one thread and a 28 ms estimated p99 at four threads. These are container-storage measurements from one current x86-64 host, not production-scale or cross-architecture performance claims.

If accountability can be expressed as a type invariant, its construction-time correctness becomes a property that compilers can check and proof assistants can verify, giving auditors machine-checked evidence that complements rather than replaces manual review, deployment controls, and legal assessment. BTV therefore prevents a class of unevidenced verdict constructions inside a defined type perimeter. It does not prove that every consequential action in a polyglot system is routed through that perimeter.

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